

NEWPORT STATE, RI, USA

WMO: 725079

Lat: 41.533N

Lon: 71.283W

Elev: 172

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14787

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	14.1	-7.8	3.6	11.9	-3.1	4.7	17.0	27.1	27.7	24.9	26.1	13.3	300	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	84.5	72.8	81.9	71.4	79.7	70.3	75.7	81.1	74.3	78.9	73.0	76.9	7.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	127.1	78.2	72.8	122.2	76.7	71.9	118.6	75.6	39.1	81.1	37.8	78.8	36.7	77.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.8	19.6	17.6	DB	3.6	90.2	5.7	3.4	-0.5	92.6	-3.8	94.6	-7.0	96.6	-11.2	99.0	
			WB	2.0	78.1	5.4	1.8	-1.9	79.4	-5.1	80.4	-8.2	81.4	-12.1	82.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	30.6	32.1	38.0	47.4	56.5	65.4	71.7	71.2	65.0	54.7	45.4	36.6
	DBStd	15.97	9.74	8.28	7.61	6.16	6.03	5.72	4.54	4.46	5.57	7.09	7.74	8.51
	HDD50	2247	603	501	377	120	9	0	0	0	0	36	179	422
	HDD65	5563	1068	920	838	530	272	64	4	5	68	325	589	880
	CDD50	2732	1	1	4	41	211	461	674	658	451	183	40	7
	CDD65	571	0	0	0	1	9	74	213	198	69	7	0	0
	CDH74	2623	0	0	0	9	62	348	1088	918	185	13	0	0
	CDH80	489	0	0	0	2	11	65	239	153	18	1	0	0
Wind	WSAvg	8.1	9.4	9.3	9.7	9.0	7.9	7.0	6.4	6.3	6.9	8.0	8.7	9.0
Precipitation	PrecAvg	46.3	3.7	3.5	4.7	4.1	3.5	3.6	3.3	3.8	4.0	3.9	3.9	4.4
	PrecMax	56.1	6.2	6.5	15.2	8.0	7.4	10.7	10.1	7.8	9.3	13.0	6.7	7.2
	PrecMin	37.1	1.7	1.8	0.3	1.5	1.1	0.0	1.0	0.8	0.8	0.4	0.7	1.3
	PrecStd	5.3	1.2	1.3	3.0	1.7	1.3	2.6	1.8	1.8	1.9	2.6	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	54.4	61.8	72.6	81.0	86.0	90.1	87.8	82.2	75.4	64.6	59.5
		MCWB	52.3	49.4	51.1	57.7	66.7	72.2	76.2	75.1	71.4	66.2	60.2	56.4
	2%	DB	51.3	50.2	55.3	64.5	74.2	81.4	85.5	83.8	78.7	70.4	62.6	56.4
		MCWB	49.2	47.0	47.7	53.3	63.3	69.5	73.6	72.7	69.1	65.1	58.6	53.8
	5%	DB	47.9	47.3	51.9	60.6	70.1	78.1	82.2	81.3	75.5	67.9	60.3	53.7
		MCWB	44.9	43.8	46.3	51.1	60.7	68.0	71.9	71.7	68.5	63.6	56.6	51.8
	10%	DB	44.6	44.6	48.5	56.8	66.2	74.8	79.8	79.1	73.4	65.5	57.4	50.3
		MCWB	41.5	40.6	44.0	49.2	58.5	66.0	70.9	70.6	67.9	61.6	54.1	47.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	51.6	54.4	60.9	69.1	74.5	78.1	77.8	75.1	69.9	62.9	57.9
		MCDB	54.1	52.4	57.4	67.1	77.4	82.4	87.3	84.3	80.0	72.4	63.8	58.9
	2%	WB	49.8	48.1	50.6	56.7	65.5	72.2	75.8	75.9	72.8	67.1	60.2	54.7
		MCDB	50.8	49.6	53.1	61.3	71.5	78.9	82.0	80.6	75.6	69.3	61.7	55.9
	5%	WB	45.7	44.8	47.7	54.0	63.0	70.1	74.3	74.5	71.3	64.7	57.6	51.9
		MCDB	47.4	46.6	50.4	57.6	67.8	75.1	79.4	78.4	73.7	66.8	59.3	53.1
	10%	WB	41.7	41.0	45.1	51.5	60.5	68.3	72.9	73.1	69.8	62.5	55.0	47.8
		MCDB	44.4	44.0	47.6	54.9	64.6	72.7	77.2	76.5	72.3	64.8	57.2	50.0
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.5	13.7	14.6	14.6	14.0	13.7	13.7	14.0	13.8	13.3	12.7
		MCDBR	15.3	15.8	17.0	19.9	19.8	18.1	16.8	15.4	15.0	15.1	14.0	15.5
		MCWBR	15.0	13.9	12.1	11.3	11.1	9.7	7.9	7.6	9.2	11.0	12.7	15.9
	5% WB	MCDBR	15.4	15.8	14.9	16.0	17.1	15.2	13.6	12.2	11.3	13.3	12.6	15.2
		MCWBR	16.1	15.5	13.0	11.7	11.0	9.2	7.4	7.0	8.2	11.6	13.0	16.5
Clear-Sky Solar Irradiance	taub	0.304	0.312	0.335	0.372	0.413	0.457	0.482	0.440	0.373	0.343	0.314	0.302	
	taud	2.509	2.487	2.462	2.393	2.309	2.223	2.161	2.274	2.434	2.525	2.563	2.561	
	Ebn at Noon	271	286	290	285	274	261	253	261	273	271	266	263	
	Edh at Noon	23	27	31	36	41	44	47	41	32	26	22	21	
All-Sky Solar Radiation	RadAvg	592	871	1192	1538	1705	1862	1912	1718	1400	948	647	497	
	RadStd	46	86	87	164	164	184	101	90	102	86	46	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+1.42	+1.31	+0.86	N/A	N/A	+216	+137
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171	+85

Nomenclature: See separate page